Multi-Campus Enterprise Network Design

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Declaration

I hereby declare that the project report entitled **“ Multi-Campus Enterprise Network Design”**, submitted in partial fulfilment of the requirements for the degree of Master in Computer Application, is my original work and has not been submitted previously to any other institution or university for the award of any degree or diploma.

The work presented in this project has been carried out entirely by me under the valuable guidance of **Ms. Damanpreet Kaur**, who provided continuous supervision, encouragement, and expert advice throughout the development process.

I affirm that all the network design, configuration, implementation, and testing performed in this project are the result of my own understanding, research, and practical application using Cisco Packet Tracer.

References to external resources, networking concepts, and configuration guidelines have been appropriately acknowledged wherever applicable.

I take full responsibility for the authenticity and originality of this work and for any discrepancies, if found, within this project report. This project is a true representation of the knowledge and practical networking skills I have gained during the course of my MCA program and through independent exploration of computer networking concepts and simulation tools.

Acknowledgment

It is with great gratitude and appreciation that I acknowledge the contributions and guidance of all those who made the successful completion of this project possible.

First and foremost, I express my heartfelt thanks to **Ms. Damanpreet Kaur**, whose mentorship, patience, and continuous encouragement were invaluable throughout the project. Her profound knowledge of computer networking concepts, including routing, switching, VLAN configuration, and network design principles, guided me in developing this project with technical accuracy and practical relevance. Her emphasis on real-world applications helped me bridge the gap between theoretical networking concepts and their practical implementation using simulation tools.

I would also like to extend my sincere thanks to the faculty members of the Computer Applications department, who have imparted essential knowledge in the fields of computer networks, data communication, and system administration. Their teachings provided the strong foundation required to design, configure, and analyze complex network infrastructures effectively.

My heartfelt gratitude goes to my family and friends, who provided immense support, motivation, and understanding throughout the duration of the project. Their encouragement during long hours of network configuration, troubleshooting, and report preparation helped me remain focused and determined.

I also acknowledge the learning resources and communities related to Cisco Packet Tracer, including tutorials and documentation that provided valuable insights into network design, configuration techniques, and troubleshooting strategies.

Lastly, I recognize that this project not only strengthened my technical capabilities in computer networking but also enhanced my understanding of enterprise network design, VLAN segmentation, routing mechanisms, and real-world network management practices. This experience has been highly enriching and has equipped me with practical skills applicable to modern networking environments.

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1.Introduction

1.1 Background

In today's rapidly evolving digital environment, computer networks have become one of the most critical components of any organization. Educational institutions, corporate enterprises, and industries rely heavily on efficient and secure network infrastructures to enable seamless communication, data sharing, and resource management across multiple departments and locations.

Traditional networks, which were once simple and localized, have now transformed into complex, multi-layered architectures that require proper planning, segmentation, and management. Technologies such as Virtual Local Area Networks (VLANs), routing protocols, and hierarchical network design play a vital role in ensuring scalability, performance, and security within modern networks.

The **Multi-Campus Enterprise Network Design project** was conceptualized as a comprehensive simulation system that provides a structured approach to designing and managing a large-scale network. The project represents a real-world scenario where a main campus and a branch campus are interconnected through routers and cloud infrastructure, enabling communication between various departments such as Admin, HR, Finance, IT, and Student Labs.

This project leverages Cisco Packet Tracer, a powerful network simulation tool, to design, configure, and analyze the network. The simulation environment allows the implementation of routing, VLAN segmentation, and connectivity testing, helping to visualize how data flows across different parts of the network and how various networking components interact with each other.

1.2 Project Objective

The primary objective of this project is to design and implement a structured and scalable multi-campus network that ensures efficient communication, proper segmentation, and reliable connectivity between all network devices and departments.

The specific goals include:

- To design a multi-campus network architecture consisting of a Main Campus and a Branch Campus
- To implement VLANs for different departments such as Admin, HR, Finance, Business, IT, and Student Labs
- To configure inter-VLAN routing using Layer 3 switches and routers
- To establish communication between campuses using WAN links
- To simulate cloud connectivity for external services such as email servers
- To ensure efficient data transmission and network performance through proper configuration
- To strengthen practical knowledge of networking concepts such as routing, switching, VLANs, and IP addressing

1.3 Scope

The scope of this project extends to the design and simulation of a multi-campus enterprise network that provides structured communication and efficient data flow across different departments and locations.

The network focuses on:

- Logical network design — creating a structured topology with routers, switches, and end devices
- Network segmentation — dividing the network into multiple VLANs for better security and performance
- Inter-network communication — enabling communication between different VLANs and campuses
- Operational visibility — understanding how data flows within and between networks

The project does not include real-time hardware implementation or advanced network security mechanisms such as firewalls and intrusion detection systems. Instead, it focuses on simulation-based network design and configuration using static topology. However, the design framework allows for future enhancements such as integration with real hardware devices, dynamic routing protocols, and advanced security features.

2. Technology Stack

Component	Tool / Technology	Description
Network Simulation Tool	Cisco Packet Tracer	Used to design, configure, and simulate the entire network topology including routers, switches, and end devices
Networking Devices	Routers (Cisco 2911), Switches (2960, 3560)	Core hardware components used for routing, switching, and inter-VLAN communication
Communication Protocol	TCP/IP	Fundamental protocol suite used for communication between devices in the network
Network Segmentation	VLAN (Virtual LAN)	Used to logically divide the network into multiple departments for improved performance and security
Routing Mechanism	Static Routing / Inter-VLAN Routing	Enables communication between different networks and VLANs
Transmission Media	Ethernet (Copper Cables)	Used to establish physical connections between devices
IP Addressing Scheme	IPv4 Addressing	Used to assign unique addresses to each device in the network
Server	Email Server (Cloud)	Simulates external communication and network services

Component	Tool / Technology	Description
Operating Environment	Windows 10	Platform used to run the simulation software
Output Format	.pkt File	Packet Tracer project file used for saving and sharing the network design

Technology Description:

The core technology used in this project is Cisco Packet Tracer, which provides a virtual environment for designing and testing network configurations without the need for physical hardware. It supports a wide range of networking devices and protocols, making it ideal for educational and simulation purposes.

Routers and switches form the backbone of the network, where routers are responsible for inter-network communication and switches manage communication within local networks. VLAN technology is used to segment the network into different departments, ensuring better security, reduced broadcast traffic, and improved performance.

The TCP/IP protocol suite is used to enable communication between devices, while IPv4 addressing ensures that each device has a unique identity within the network. Routing techniques, including inter-VLAN routing, allow seamless communication across different network segments.

Additionally, the use of a cloud-based email server demonstrates how external services can be integrated into a network environment. Overall, the technology stack ensures that the network is scalable, efficient, and aligned with real-world enterprise networking practices.

3.Project Structure Overview

The Multi-Campus Enterprise Network is designed to simulate a real-world organizational network in a structured, scalable, and efficient manner. It provides a comprehensive view of how different departments across multiple campuses are interconnected using networking technologies such as VLANs, routing, and switching. The network highlights key aspects such as segmentation, communication flow, and centralized management, ensuring efficient data transfer and operational reliability.

The project follows a structured approach involving:

1. **Network Design & Planning** – Designing the overall topology including Main Campus and Branch Campus, selecting appropriate networking devices such as routers, switches, and end devices, and defining IP addressing schemes and VLAN structures for different departments.
2. **Configuration & Implementation** – Configuring VLANs, assigning ports, enabling trunk links between switches, setting up inter-VLAN routing using Layer 3 switches and routers, and establishing WAN connectivity between campuses along with cloud integration for external services.
3. **Testing & Analysis** – Verifying network connectivity using tools such as ping, analyzing communication between VLANs and campuses, identifying and resolving configuration issues, and ensuring that the network performs efficiently under the designed topology.

The network provides a holistic view of enterprise-level communication, from internal departmental connectivity to inter-campus data exchange, thereby enabling efficient resource sharing and centralized network management.

Dashboard =>

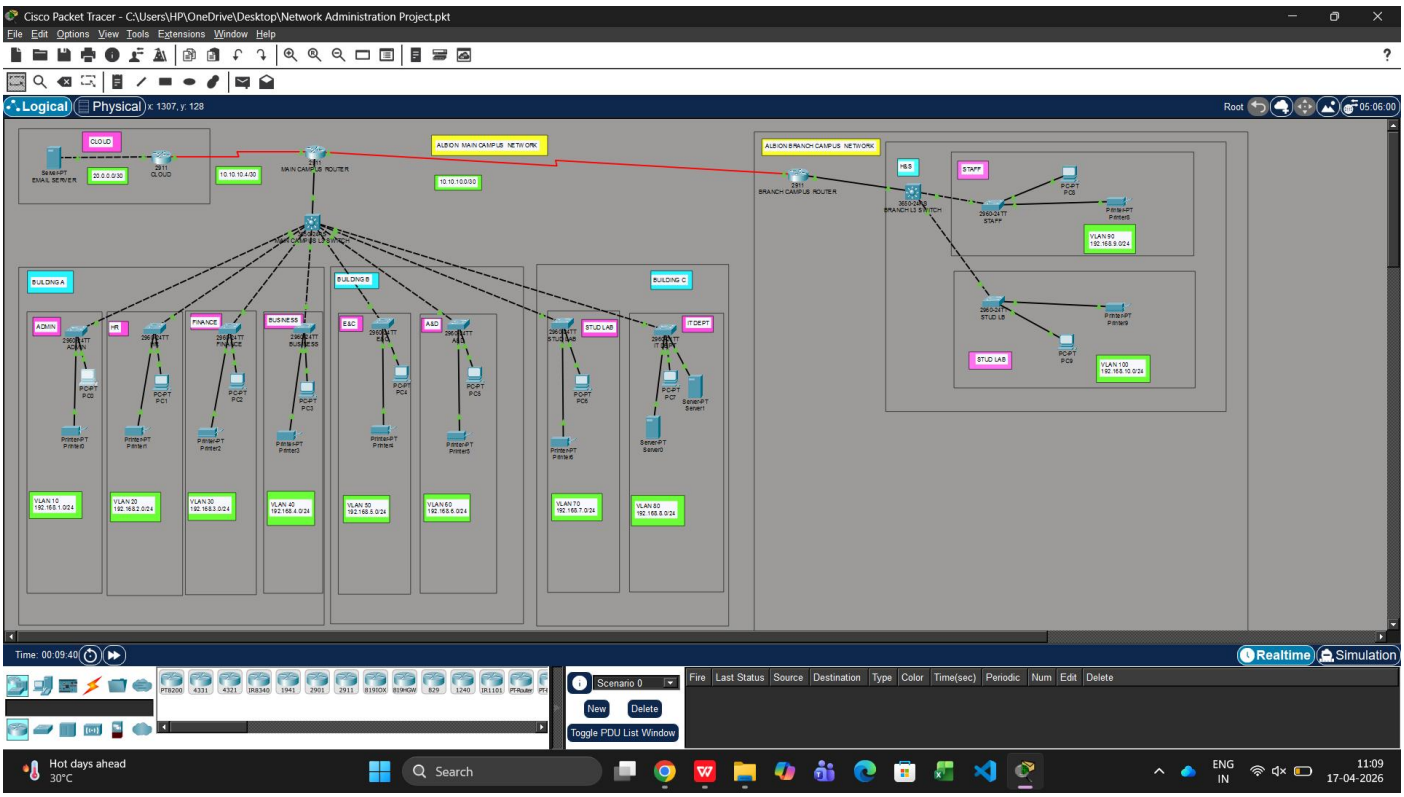

Fig:Campus Network

4.Network Components Description

The network used in this project represents a simulated enterprise infrastructure designed using Cisco Packet Tracer. Instead of a traditional dataset, this project consists of interconnected networking devices, IP configurations, and VLAN-based segmentation that together form a structured multi-campus network system.

The network includes multiple components such as routers, switches, end devices, and servers, each configured with specific roles and parameters to enable communication across departments and campuses.

Key Components:

- **Routers:**
  Devices used to connect different networks and enable communication between the Main Campus and Branch Campus through WAN links.
- **Switches:**
  Used to connect multiple devices within a Local Area Network (LAN) and to manage VLAN-based segmentation.
- **End Devices (PCs):**
  Represent users in different departments such as Admin, HR, Finance, IT, and Student Labs.

- **VLANs (Virtual LANs):**
  Logical segmentation of the network into departments such as Admin, HR, Finance, Business, IT, and Labs to improve performance and security.
- **IP Addressing Scheme:**
  Each VLAN is assigned a unique subnet (e.g., 192.168.1.0/24, 192.168.2.0/24), ensuring proper identification and communication between devices.
- **WAN Links:**
  Used to connect the Main Campus and Branch Campus networks using a separate subnet (e.g., 10.10.10.0/30).
- **Cloud / Server:**
  Represents external network services such as an Email Server, allowing communication beyond the local network.

Network Size:

The network consists of multiple VLANs (10–100), several switches, routers, and end devices distributed across two campuses (Main and Branch), simulating a medium-scale enterprise network.

Data Type:

The project includes a combination of:

- **Configuration Data:** VLAN IDs, IP addresses, routing tables
- **Logical Data:** Network topology and connectivity
- **Operational Data:** Communication between devices (ping results, routing paths)

Purpose:

The purpose of this network design is to:

- Enable structured communication between multiple departments
- Implement network segmentation using VLANs
- Ensure connectivity between campuses through routing
- Simulate real-world enterprise network architecture
- Analyze how data flows across different network segments

5.Network Design & Structure

The Multi-Campus Enterprise Network is structured into multiple interconnected layers and segments integrated into one comprehensive network topology. It provides a unified view of communication across different departments, buildings, and campuses, ensuring efficient data flow and centralized network management.

Network Components

1. Core Layer (Top-Level Connectivity):

- Represents the backbone of the network.
- Includes routers responsible for connecting the Main Campus, Branch Campus, and external cloud network.
- Ensures high-level routing and communication between different networks.

2. Distribution Layer (Inter-VLAN Routing):

- Consists of Layer 3 switches used for routing between VLANs.
- Handles traffic between different departments within the campus.
- Acts as an intermediary between core and access layers.

3. Access Layer (End Device Connectivity):

- Includes Layer 2 switches connecting end devices such as PCs.
- Assigns VLANs to different ports based on departments.
- Provides direct network access to users.

4. Department-wise VLAN Structure:

- Network is divided into multiple VLANs such as Admin, HR, Finance, Business, IT, and Student Labs.
- Each VLAN has a separate subnet (e.g., 192.168.x.0/24).
- Ensures logical separation, security, and reduced broadcast traffic.

5. Inter-Campus Connectivity:

- WAN links connect the Main Campus and Branch Campus using routers.
- Enables communication between geographically separated networks.
- Uses dedicated IP ranges for point-to-point connections.

6. Cloud and External Services:

- A cloud network is integrated to simulate external services such as an Email Server.
- Allows devices within the network to communicate with external systems.
- Represents real-world internet connectivity.

7. Network Monitoring and Testing:

- Connectivity is verified using tools such as ping and simulation mode in Cisco Packet Tracer.
- Helps in analyzing packet flow and identifying network issues.

6.Key Functionalities and Network Analysis

1. Department-wise Network Segmentation:

The network design reveals that different departments such as Admin, HR, Finance, IT, and Student Labs are separated using VLANs. This segmentation ensures that each department operates within its own logical network, reducing unnecessary traffic and improving overall network efficiency.

2. Device-Based Communication Analysis:

Devices across different VLANs are able to communicate through inter-VLAN routing. This indicates that the routing configuration is properly implemented, allowing seamless data exchange between users in different departments.

3. Campus-wise Connectivity Trend:

The network demonstrates strong connectivity between the Main Campus and Branch Campus. Communication across campuses is successfully achieved through WAN links, ensuring that geographically separated networks function as a single integrated system.

4. IP Addressing and Network Organization:

The structured IP addressing scheme shows that each department has a unique subnet. Networks with properly assigned IP ranges experience better organization, easier troubleshooting, and reduced chances of IP conflicts.

5. Network Traffic and Performance Behavior:

Departments connected through properly configured VLANs and trunk links experience efficient data transmission. Misconfigured or overloaded segments can lead to communication delays, highlighting the importance of correct VLAN and trunk configuration.

6. Network Reliability and Stability:

The overall network demonstrates stable performance, where all devices can communicate without packet loss. Proper routing, VLAN configuration, and connectivity ensure reliable network operation, which is essential for real-world enterprise environments.

7.Testing and Validation

Testing and validation were carried out to ensure that the network operates efficiently, maintains proper connectivity, and delivers accurate communication between all devices across different VLANs and campuses.

Testing Procedures:

1. Configuration Validation:
All network configurations, including VLAN IDs, IP addressing, subnet masks, and routing settings, were verified to ensure correct implementation across routers and switches.

2. Connectivity Testing:
Ping tests were performed between devices within the same VLAN and across different VLANs to ensure proper communication and successful inter-VLAN routing.

3. Inter-Campus Communication Testing:
Connectivity between the Main Campus and Branch Campus was tested using WAN links to confirm that devices from both campuses can communicate without interruption.

4. Performance Testing:
The network was tested under simulation mode in Cisco Packet Tracer to analyze packet flow, detect delays, and ensure efficient data transmission across the network.

5. Troubleshooting and Validation:
Common issues such as VLAN mismatch, incorrect IP configuration, and trunking errors were identified and resolved to ensure smooth network operation.

All tests confirmed that the network is functioning correctly, with reliable communication, proper configuration, and efficient performance suitable for a real-world enterprise environment.

8. Limitations and Challenges

Limitations:

- Simulation-based project
- No real-time traffic handling
- Limited security implementation

Challenges:

- Managing multiple VLANs (10–100)
- IP addressing accuracy
- Routing between campuses
- VLAN mismatch errors

9. Future Enhancements

1. Add **Firewall & Security (ACLs)**
2. Implement **Dynamic Routing (OSPF/EIGRP)**
3. Add **Wi-Fi network (Access Points)**
4. Cloud integration with real servers
5. Network monitoring system

10.Conclusion

The Multi-Campus Enterprise Network project successfully demonstrates a real-world network architecture using Cisco Packet Tracer.

It showcases:

- VLAN segmentation
- Inter-VLAN routing
- Multi-campus connectivity
- Cloud integration

This project enhanced my practical knowledge of networking and prepared me for real-world network design and administration.

11.References

1. Cisco Networking Academy
2. Cisco Packet Tracer Documentation
3. Computer Networking Books
4. Online Networking Tutorials